

ECN 712 Fall 2025
EE Schlee Problem Set III
Due Thursday, 11 September

1. Consider these two utility representations on the consumption set $X = \mathbb{R}_+^2$.
 - (i) $u(x_1, x_2) = \min\{\frac{x_1}{\alpha}, \frac{x_2}{\beta}\}$ where $\alpha, \beta > 0$ and (ii) $u(x_1, x_2) = ax_1 + bx_2$ where $a, b > 0$.
 - (a) Sketch a few indifference curves for each utility function. In each case, write down the demand, the indirect utility, the expenditure function, and the compensated demand.
[Notes: 1. I ask you simply to “write down” the functions; you needn’t show how you got them on paper, but you *should* be able to articulate to yourself a rigorous justification for your conclusion.
2. *Think economically*: be alert for shortcuts to avoid needless repetition.]
 - (b) Verify that each expenditure function is concave in (p_1, p_2) . Is it differentiable? If not, what preference property accounts for the lack of differentiability?
 - (c) Verify that each indirect utility function is quasiconvex in (p_1, p_2) .
2. MWG, 3.G.6. You should interpret these equations as holding for all positive price-wealth pairs such that $(x_1, x_2) \geq 0$ and $p_1x_1 + p_2x_2 \leq w$ and assume that preferences are represented by a *continuous, monotone, and quasiconcave* function $u : \mathbb{R}_+^3 \rightarrow \mathbb{R}$ throughout. Work parts (a) and (b) as written. Replace parts (c) - (e) by these slight modifications.
 - (c) Determine what must be true about α, β , and γ if $\delta > 0$.For the rest of the problem *impose* the condition that $\delta = 0$ in addition to the conditions you derived in part (c) for α, β , and γ .
 - (d) Draw the consumer’s indifference curves in the x_1 - x_2 plane for a given level of x_3 . How do you use monotonicity or quasiconcavity here? How do the conditional preferences over (x_1, x_2) depend on the level of x_3 ?
 - (e) Determine as much as you can about u ; you can pin it down up to a monotonic transformation. Justify your conclusion.
3. In Ionia, there are two goods, bread today and bread tomorrow. The poet Sappho is endowed with $\omega_1 > 0$ loaves of bread today and $\omega_2 > 0$ loaves tomorrow. The price *today* of bread to be available at date t is $p_t > 0, t = 1, 2$. Her preferences over consumption plans (c_1, c_2) in $\mathbb{R}_+^2$ are represented by a *continuous, locally nonsatiated, and strictly quasiconcave* function u , her (ordinary) demand is single-valued and differentiable in (p, w) , and her indirect utility V is C^1 with $\partial V(p, w)/\partial w > 0$ for all $(p, w) \gg 0$. (These assumptions imply that her compensated demand is differentiable in p .)
 - (a) What properties does her indirect utility function have as a function of prices? Prove it. Is there a version of Roy’s identity for her demands?
 - (b) Derive her Slutsky equation describing the effect of an increase in the price of bread today on today’s consumption. (Suggestion: insert her price-dependent wealth into the “ w ” argument of her ordinary demands; differentiate, then appeal to the Slutsky equation we derived together.)

- (c) What is the interest rate in this economy? (If you have trouble here, consult Chapter 2 of Debreu's *Theory of Value*.) Suppose that "consumption today" is a normal good for her. *Evaluate*: "If the rate of interest rises then saving, $\omega_1 - c_1$, must rise." Prove it if true, give an explanation or (possibly-graphical) counterexample if false.
- (d) Suppose that there are $T \geq 2$ dates and a single physical good, and let Ω be a nonempty compact subset of $\mathbb{R}_{++}^T$. Suppose Sappho first chooses an endowment point $\omega \in \Omega$, taking consumption prices $p \in \mathbb{R}_{++}^T$ as given.¹ After choosing $\omega \in \Omega$, she chooses a consumption plan c in her budget set, $\{c \in \mathbb{R}_+^T \mid p \cdot c \leq p \cdot \omega\}$. Do you need to know her preferences over consumption plans (beyond the maintained italicized assumptions) to determine her endowment choice? Explain. (Hint: if you have trouble, proceed to the next sentence; a picture for the $T = 2$ case might help you see an argument for the general case.)
- For the special case of $T = 2$, illustrate in a diagram both the investment choice (out of a constraint set Ω that you depict graphically) and the ultimate consumption choice.

The next part you should answer *only if* you have the time. Otherwise you should return to it before the midterm. (For what it is worth, as far as your score is concerned the only thing that this part will do is possibly change a $\checkmark$ to a $\checkmark+$, and you can earn a $\checkmark+$ without touching this part.)

- (e) $\star$ In part (a) you wrote down a version of Roy's identity for this problem. (i) Do the assumptions of the classical Envelope Theorem necessarily hold here? Explain. [Hint: No.] (ii) Returning to the standard consumer's problem with price-independent wealth, *justify* Roy's Identity,

$$-\frac{\partial V / \partial p_\ell}{\partial V / \partial w}(p, w) = d_\ell(p, w)$$

for $\ell = 1, \dots, L$, from the preference assumptions of this problem, making free use of these facts.

$$V(p, e(p, \bar{u})) = \bar{u}, \quad (1)$$

$$e(p, V(p, w)) = w, \quad (2)$$

and

$$h(p, \bar{u}) = d(p, e(p, \bar{u})), \quad (3)$$

for every $(p, w) \gg 0$ and every $\bar{u} \in \text{Range}(u)$ with $\bar{u} > u(0)$, where V is the indirect utility function, e is the expenditure function, d the ordinary, and h the compensated demand. If you use a theorem from calculus, please be clear about how you are using it. Which of the maintained assumptions of this problem can you dispense with in establishing Roy's identity? [Hint: more than one is either redundant or not needed.]

Quite apart from their uses in this problem, you would do well to think about what equations (1) - (3) assert. Try writing down a sentence for each, the more concise and to-the-point the better.

¹From her viewpoint, this endowment choice can be viewed as an *investment* decision, for example, lowering her endowment today to increase it tomorrow.